

Hand Evaluator and Probability Checker for Texas Hold'em: pypokerprob

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Abstract

This paper introduces the python library pypokerprob. The purpose of this library is to allow you to evaluate your cards dealt during a game of Texas Hold'em; with the goal of mathematically modeling the different possible hands you can create with those cards and the probabilities of getting these different hands. In addition to expressing the efficient functionality of the library, we will be covering an example that will explicitly cover the calculation of the probability of getting different types of hands.

1 Introduction

There are many different variations of poker however our code works on Texas Hold'em which has been around since the 19th century [1] and is one of the most popular card games played in casinos. The game consists of each player having 2 cards in their hands with 5 community cards which are revealed throughout the game and the aim is to get the highest possible hand. Pypokerprob is used to work out the probability of getting a certain hand as shown below:

```
>>> hand = [[4, 'spades'], [5, 'hearts'], [3, 'diamonds'], [12, 'hearts'], [4, 'hearts']]

>>> evaluated_know_cards_hand = pypokerprob.evaluate_all_known_cards(hand)

>>> pypokerprob.probability(evaluated_know_cards_hand)

{'one_pair': 0.45790934320074006,

 'two_pair': 0.36632747456059206,

 'three_of_a_kind': 0.0666049953746531,

 'straight': 0.041628122109158186,

 'flush': 0.041628122109158186,

 'full_house': 0.02497687326549491,

 'four_of_a_kind': 0.0009250693802035153,

 'straight_flush': 0.0,

 'royal_flush': 0.0}
```

This code runs all possible combinations of cards that are yet to be drawn using combinatorics to run all the possible hands that could be produced. The probability is worked out using the evaluate function and dividing every value by all possible combinations.

2 Statement of Need

The mathematical representation of hand evaluators and probability checkers of poker on a whole and it's variant games like Texas Hold'em is highly valued within the research community. As it's analysis has played a key role in the area of Artificial Intelligence (AI) research, due to the complexity it possess by it's association with stochasticity, imperfect information, and chance events [3]. Additionally, this high association Texas Hold'em has with imperfect information has played an important role in a variety of real world applications. Some of these applications are military decision-making, trading, business management, and many more [2]. Although the purpose and use of the pypokerprob library may seem pretty hard to grasp, as nothing more than a smart way of seeing if you have a good chance at winning in a game of Texas Hold'em. It's true purpose is much more. To us it's purpose is to be a python library that is efficient, easily accessible, and systematically structured such that it can be used alongside other scientific tools. Thus, with the hope of allowing it to aid in the different fields spoken about above and in the many other fields to come in the future.

At it's core the software we have created is a hand evaluator; that is able to tell you the different possible hands that can be formed from your 'hole' cards and the best hands that can be formed from your 'hole' and 'flop' cards. Furthermore, it is a probability checker; which tells you the different probabilities of getting the different types of hands possible in Texas Hold'em from given cards. But, as covered above, the software we have created has much more to it than what meets the eye. Yes, it's main functionality is to evaluate given cards, discover the different hands that are possible to get with those cards, and enumerate the probabilities of getting those specific hands. But it's purpose to us is much more and it's potential alongside other scientific tools. We hope will help in further research and development as we progress into the future.

3 Conclusion

Throughout the paper we have demonstrated the use of pypokerprob and how it can be used to help with making real world decisions like trading and business management. However the library could be improved by making functions like the evaluator function more efficient so it can run quicker for hands made with fewer known cards. The library could further improved by adding a feature which could calculate the optimal bet and add money to make it more usable in actual poker games. Are library is written in a modular way, documented and tested.

It can be downloaded by:

1. Open the command terminal, abbreviated as CMD on the windows search bar.
2. If you have th folder already downloaded on your device, you can put the path pyproject.toml or "C:-First-year-project-Poker-Harris-Dylan-Jared-James.toml" into your command terminal.
3. Once you have found the folder required, you can use the command: "pip install -e ."

The source code is available at: <https://github.com/Lohaza/Uni-First-year-project-Poker-Harris-Dylan-Jared-James/blob/main/>

References

- [1] Valtteri Haarahiltunen. The terminology of poker: an overview of translated terms in a specialized field. Master's thesis, Itä-Suomen yliopisto, 2017.
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- [3] Jonathan Rubin and Ian Watson. Computer poker: A review. *Artificial Intelligence*, 175(5):958–987, 2011. Special Review Issue.